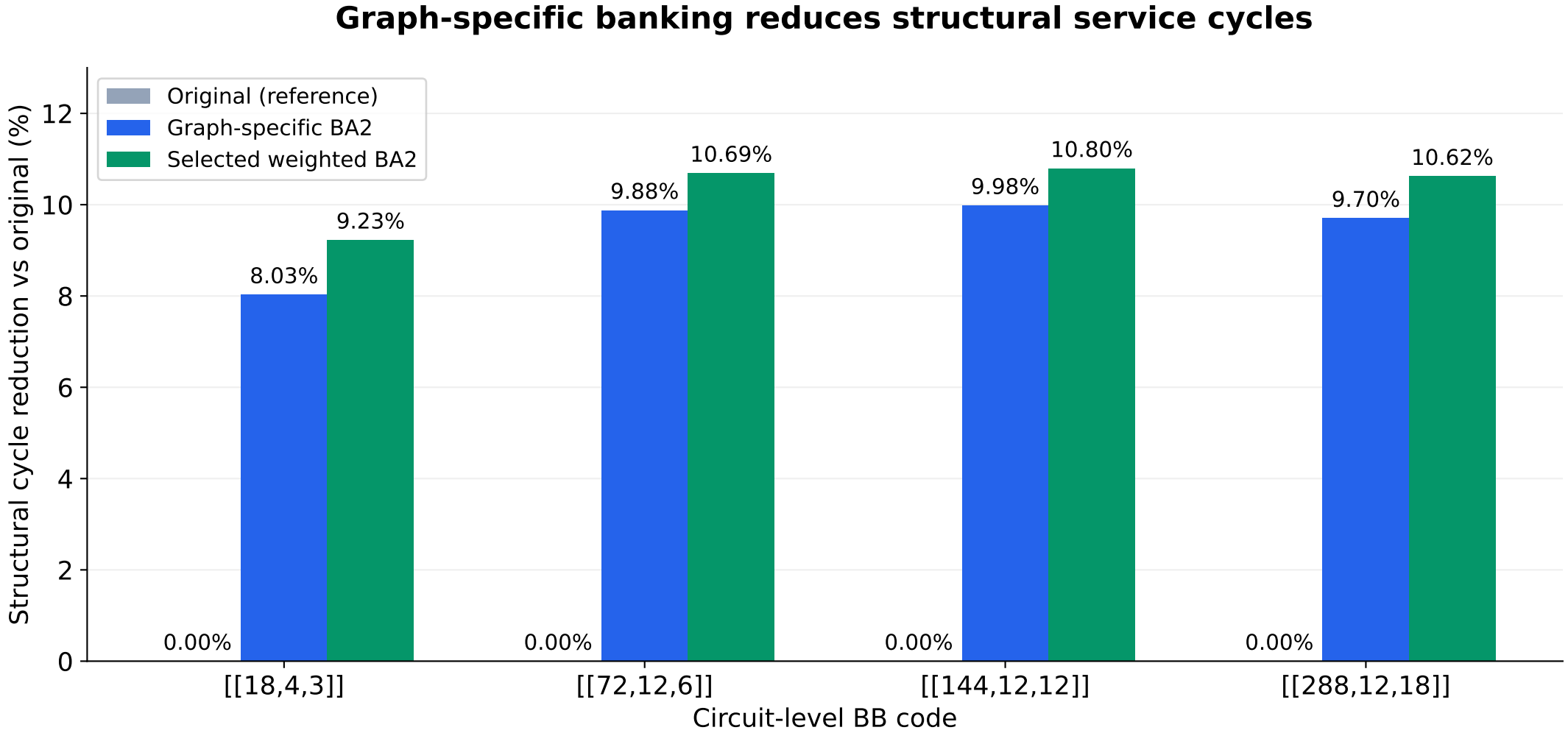

P = 4; one full iteration + one relay initialization, identical structural phase accounting for all layouts.
Verified four-shot pilot layouts (2/2 iterations, R2/S1); profile values are seed-independent. Not FPGA/RTL measurements.